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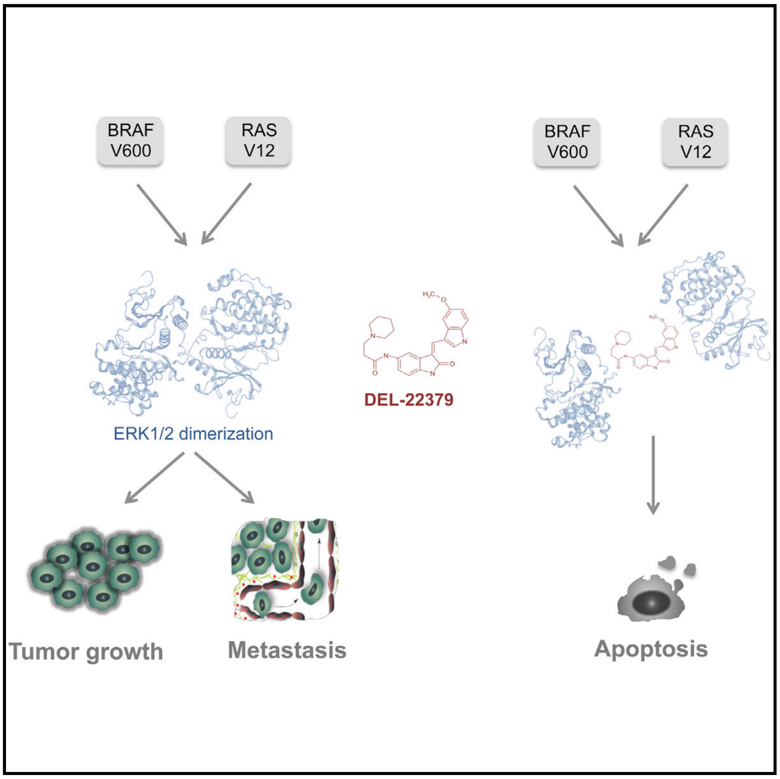
Small Molecule Inhibition of ERK Dimerization Prevents Tumorigenesis by RAS-ERK Pathway Oncogenes

Graphical Abstract



Highlights

- DEL-22379 inhibits ERK dimerization without affecting its phosphorylation.
- DEL-22379 blocks proliferation of tumor cells harboring RAS-ERK pathway oncogenes.
- In animal tumor models, DEL-22379 induces apoptosis, preventing tumor progression.
- DEL-22379 is unaffected by resistance mechanisms that hamper BRAF/MEK inhibitors.

Authors

Ana Herrero, Adán Pinto, Paula Colón-Bolea, ..., Héctor G. Palmer, Adam Hurlstone, Piero Crespo

Correspondence

crespop@unican.es

In Brief

Herrero et al. identify a small molecule inhibitor of ERK dimerization that impedes the growth of tumor cells dependent on a hyperactivated RAS-ERK pathway. Importantly, the antitumor effect of this compound in cells is not affected by the reported resistance mechanisms for the current BRAF and MEK inhibitors.

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Small Molecule Inhibition of ERK Dimerization Prevents Tumorigenesis by RAS-ERK Pathway Oncogenes

Ana Herrero,[1,12] Adán Pinto,[1,12] Paula Colón-Bolea,[1,12] Berta Casar,[1,12] Mary Jones,[2] Lorena Agudo-Ibáñez,[1] Rebeca Vidal,[1,3] Stephan P. Tenbaum,[4] Paolo Nuciforo,[4] Elsa M. Valdizán,[1,3] Zoltan Horvath,[5] Laszlo Orfi,[5,6] Antonio Pineda-Lucena,[7] Emilie Bony,[8] Gyorgy Keri,[5,9] Germán Rivas,[10] Angel Pazos,[1,3] Rafael Gozalbes,[11] Héctor G. Palmer,[4] Adam Hurlstone,[3] and Piero Crespo[1,*]

[1]Instituto de Biomedicina y Biotecnología de Cantabria (IBBTEC), Consejo Superior de Investigaciones Científicas (CSIC), Universidad de Cantabria, Santander 39011, Spain
[2]Faculty of Life Sciences, University of Manchester, Manchester M13 9PL, UK
[3]Departamento de Fisiología y Farmacología, Centro de Investigación Biomédica en Red de Salud Mental (CIBERSAM), Instituto de Salud Carlos III Universidad de Cantabria, Santander 39011, Spain
[4]Stem Cells and Cancer Laboratory, Translational Research Program, Vall d'Hebron Institute of Oncology (VHIO), Barcelona 08035, Spain
[5]Vichem Chemie Research Ltd., 1022 Budapest, Hungary
[6]Department of Pharmaceutical Chemistry, Semmelweis University, 1092 Budapest, Hungary
[7]Centro de Investigación Príncipe Felipe, 46012 Valencia, Spain
[8]Pharmacognosy Research Group, Louvain Drug Research Institute, Université Catholique de Louvain, 1200 Bruxelles, Belgium
[9]MTA-SE Pathobiochemistry Research Group, Department of Medical Chemistry, Semmelweis University, 1092 Budapest, Hungary
[10]Centro de Investigaciones Biológicas, Consejo Superior de Investigaciones Científicas, 28040 Madrid, Spain
[11]ProtoQSAR S.L., 46008 Valencia, Spain
[12]Co-first author
*Correspondence: crespop@unican.es
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SUMMARY

Nearly 50% of human malignancies exhibit unregulated RAS-ERK signaling; inhibiting it is a valid strategy for antineoplastic intervention. Upon activation, ERK dimerize, which is essential for ERK extranuclear, but not for nuclear, signaling. Here, we describe a small molecule inhibitor for ERK dimerization that, without affecting ERK phosphorylation, forestalls tumorigenesis driven by RAS-ERK pathway oncogenes. This compound is unaffected by resistance mechanisms that hamper classical RAS-ERK pathway inhibitors. Thus, ERK dimerization inhibitors provide the proof of principle for two understudied concepts in cancer therapy: (1) the blockade of sub-localization-specific sub-signals, rather than total signals, as a means of impeding oncogenic RAS-ERK signaling and (2) targeting regulatory protein-protein interactions, rather than catalytic activities, as an approach for producing effective antitumor agents.

INTRODUCTION

The signaling pathway connecting RAS guanosine triphosphatases and mitogen-activated protein kinases ERK1 and ERK2 (ERK) is essential for the regulation of cellular proliferation and survival (Arozarena et al., 2011). Activating mutations in constituents of the RAS-ERK pathway, particularly in members of the RAS and RAF families, are common in neoplasia, appearing in nearly 50% of human cancers (Matallanas and Crespo, 2010). Aberrant flux through the RAS-ERK pathway plays a critical

Significance

Nearly 50% of human cancers exhibit unregulated signaling through the RAS-ERK pathway. Strategies aimed at reducing aberrant flux through this route are attractive options for antitumor therapy. Activated ERK dimerize, playing an essential role in ERK's extranuclear functions but not its nuclear functions. Here, we report on a small molecule inhibitor for ERK dimerization that impedes tumorigenesis driven by RAS-ERK pathway oncogenes and demonstrate that the blockade of sub-localization-specific sub-signals can be effective for counteracting oncogenic RAS-ERK signaling. Our results show that targeting regulatory protein-protein interactions such as dimerization, rather than catalytic activities, can be a valid approach for producing effective antitumor agents.

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role in the origin, progression, and maintenance of malignancy (Pylayeva-Gupta et al., 2011). Thus, in the past decades, academia and industry have devoted colossal efforts to find drugs whereby oncogenic RAS-ERK signaling could be curtailed.

The recent years have seen significant advances in the development of pharmacological agents directed against the kinases that populate the RAS-ERK pathway. Compounds inhibiting the kinase activities of BRAF, MEK, and ERK have yielded promising preclinical results, and some have progressed to the clinic for the treatment of some types of tumors, particularly melanoma, with substantial, though short-lived, success (Montagut and Settleman, 2009; Morris et al., 2013). However, an effective and lasting use of these molecules has been hampered by undesired side effects and the emergence of drug resistance. Hitherto, most of the molecular mechanisms leading to the acquisition of resistance to BRAF and MEK inhibitors involve reactivation of ERK signaling (Little et al., 2013). Therefore, unveiling alternative mechanisms of action whereby the RAS-ERK pathway could be targeted and that could avoid such drug-resistance processes is highly desirable.

It has long been known that ERK dimerize in response to phosphorylation (Khokhlatchev et al., 1998). In this respect, we have previously shown that ERK dimerization plays an essential role as an spatial regulator of ERK signals (Calvo et al., 2010) by specifically orchestrating ERK extranuclear, but not nuclear, signaling (Casar et al., 2008, 2009). Furthermore, by molecular biology approaches, we demonstrated that impeding ERK dimerization, and thereby ERK signals' extranuclear component, was sufficient for curtailing cellular transformation and tumor development (Casar et al., 2008), unveiling ERK dimerization as an attractive molecular target for the development of antineoplastic drugs based on an understudied concept: the disruption of regulatory protein-protein interactions. Thus, the goal of the present study was to find small molecules capable of preventing ERK dimerization.

RESULTS

Characterization of DEL-22379 as an ERK Dimerization Inhibitor

Detecting ERK dimerization by SDS-PAGE (Philipova and Whitaker, 2005) is a delicate technique with a high rate of technical failures and is therefore unsuitable for screening large numbers of compounds. Thus, we set up a different procedure: detecting ERK dimerization by native PAGE electrophoresis, which would facilitate the process in comparison to the previous assay. This methodology was validated using ERK dimer and monomer purification by high-performance liquid chromatography, analytical centrifugation of ERK monomers and dimers, and molecular biology approaches, which ascertained that the faster- and slower-migrating bands in native PAGE electrophoresis corresponded to ERK dimers and monomers, respectively (Figures S1A–S1C). A higher negative charge and dipole moment for the dimer, probably as a consequence of the embedding of charged residues upon dimerization (Table S1), may explain its enhanced electrophoretic mobility. Using this assay, several chemical libraries, amounting to ~1,100 compounds of different natures, were screened. We reasoned that the greatest chances of finding an inhibitor would be among derivatives from chemical structures with proven affinity toward kinases, so we screened the extended validation library of Vichem's Nested Chemical Library based on core structures of known, validated kinase inhibitors. We tested 650 compounds, including pyrazolo-pyrimidines, 2,6-phenyl-pyrimidines, and indanones. Therein, we identified 25 compounds, for example, #2 (Figure 1A), which inhibited epidermal growth factor (EGF)-induced ERK dimerization as a consequence of inhibiting ERK phosphorylation. Two other compounds, #1 and #3, inhibited ERK dimerization without affecting its phosphorylation; #3 was the water-soluble 3-arylidene-2-oxindole derivative DEL-22379 (Figure 1B).

Further validation for DEL-22379 as an ERK dimerization inhibitor in vivo was obtained using SDS-PAGE electrophoresis (Philipova and Whitaker, 2005) for lysates from HEK293 cells, in which ERK dimerization stimulated by EGF was abolished by treatment with the molecule (Figure 1C). Also in these cells, DEL-22379 abolished EGF-induced co-immunoprecipitation of ectopic ERK2 molecules tagged with hemagglutinin (HA) or FLAG epitopes, with an estimated half-maximal inhibitory concentration (IC_{50}) of ~0.5 μM (Figure 1D). In both assays, the ability of DEL-22379 to disrupt ERK dimerization was not related to ERK phosphorylation, which was unaffected by the compound.

To visualize how DEL-23379 affected ERK dimerization in live cells, we performed proximity ligation assays (PLAs). ERK dimerization was more prominent at the cytoplasm of EGF-stimulated HeLa cells, in agreement with our previous studies (Casar et al., 2008). Contrarily, ERK dimers were undetectable in cells previously treated with DEL-22379 (Figures 1E and S1D). The ERK dimer interface is formed from a nonhelical leucine zipper (Wilsbacher et al., 2006). Heterodimerization of the transcription factors MYC and MAX takes place by a similar interaction (Lüscher and Larsson, 1999). However, MYC/MAX heterodimerization was unaffected by DEL-22379 (Figure S1E). This demonstrated that DEL-22379 generally does not interfere with all protein-protein interactions of the leucine zipper type.

It was important to determine whether DEL-22379 prevented the formation of ERK dimers in vitro. To this end, ERK2 purified from bacteria was phosphorylated in vitro by purified, activated MEK1 ΔN EE in the presence of increasing concentrations of the compound. ERK2 dimerization was progressively inhibited with an IC_{50} of ~0.5 μM (Figure S1F), without alterations in ERK2 phosphorylation (Figure 1F). However, DEL-22379 could not disrupt preformed dimers (Figure S1G). To test whether DEL-22379 affected ERK kinase activity, we performed in vitro kinase assays, using myelin basic protein (MBP) as a substrate for ERK immunoprecipitated from EGF-stimulated cells pretreated with DEL-22379 or the MEK inhibitor U0126 as a control. It was found that ERK kinase activity was unaltered by DEL-22379 (Figure 1G). Overall, these data demonstrate that DEL-22379 is a bona fide inhibitor of ERK dimerization, but it does not affect ERK phosphorylation or ERK kinase activity.

It was of interest to know how DEL-22379 docked onto ERK. Therefore, we performed in silico docking studies, using the available structural data on ERK2 (Canagarajah et al., 1997; Wilsbacher et al., 2006) to generate a feasible docking model. It was not possible to situate the ligand on the dimerization surface. In contrast, it docked on the cleft defined by residues Pro 174, Asp 175, His 176, Asp 177, Phe 181, and Phe 329 (Figure 2A; Figures S2A–S2D); His 176 was particularly relevant, because charge

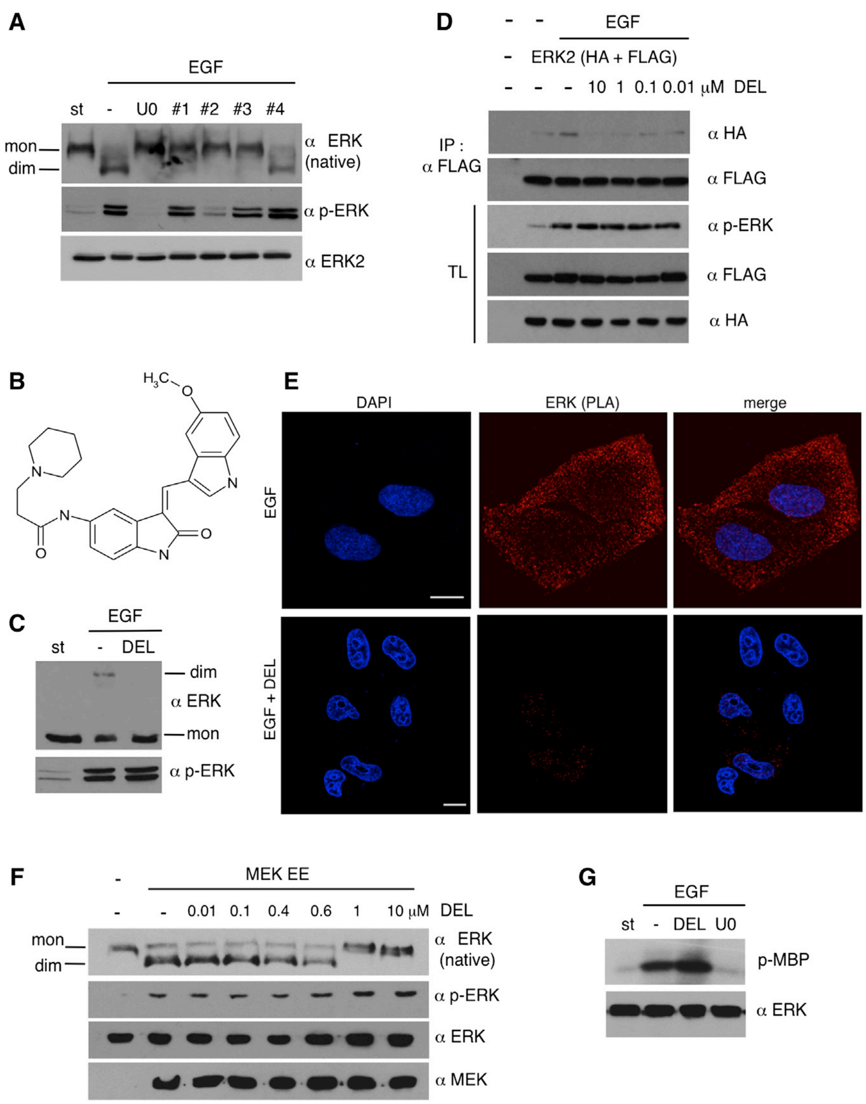

Figure 1. Identification and Validation of DEL-22379 as an ERK Dimerization Inhibitor

(A) Native PAGE of compounds that inhibit ERK dimerization without affecting ERK phosphorylation. Starved HEK293 cells were treated with the inhibitors or U0126 as the positive control (10 μM for 30 min) and stimulated with EGF (100 ng/ml for 10 min). St, starved cells.

(B) Structure of DEL-22379.

(C) ERK dimerization analyzed by SDS-PAGE in lysates from HEK293 cells treated with EGF and DEL-22379.

(D) Concentration-dependent effect of DEL-22379 on the co-immunoprecipitation of HA- and FLAG-tagged ERK2 in HEK293 cells.

(E) Effects of DEL-22379 on dimerization of HA- and FLAG-tagged ERK2 in live HeLa cells determined by PLA. Scale bars, 10 μm.

(F) DEL-22379 effect on ERK2 dimerization in vitro. Purified ERK2 was incubated with activated GST-MEK ΔN EE plus increasing concentrations of DEL-23379.

(G) DEL-22379 effect on ERK2 kinase activity, determined in ERK2 immunoprecipitates using MBP as a substrate.

See also Figure S1 and Table S1.

repulsion at this position is sufficient to disrupt ERK2 dimerization (Wilsbacher et al., 2006). This approach suggested that DEL-22379 bound to ERK2 at a groove within the dimerization interface, in agreement with its inhibitory effect on the formation of ERK2 homodimers. In order to validate experimentally this theoretical docking site, we examined DEL-22379 binding to ERK2 using microscale thermophoresis. DEL-22379 readily bound to ERK2 with a K_d estimated in the low micromolar range, though binding was detectable even at low nanomolar concentrations. Contrarily, binding was impaired to different extents for the ERK2 mutants D175A, H176L, F181A, and F239A, in which the putative binding site had been altered (Figure 2B), thereby verifying DEL-22379 docking at the in silico predicted site. These results demonstrated that DEL-22379 docked onto ERK2 within the dimerization interface, thereby explaining the mechanism whereby this compound interferes with the formation of ERK dimers.

We had previously demonstrated that ERK dimers play a role in ERKs extranuclear, but not nuclear, signaling. Using an ERK2 mutant H176E, L_4A (Khokhlatchev et al., 1998) (HL hereafter) that functions in a dominant inhibitory fashion, impeding ERK dimerization, we had shown that preventing dimerization diminished ERK cytoplasmic signaling but enhanced its nuclear functions (Casar et al., 2008). Conceivably, DEL-22379 should behave similarly. Indeed, treating HEK293 cells with DEL-22379 prevented the phosphorylation of the cytoplasmic kinase RSK1 induced by EGF treatment (Figure 2C) while enhancing the transactivation of the transcription factor ELK (Figure 2D) to levels similar to those detected in cells expressing ERK2 HL. DEL-22379-induced upregulation in ERK nuclear activity was further confirmed by the enhanced phosphorylation of the transcription factors ELK, FOS, and MYC, but not JUN, which is not an ERK substrate (Figure 2E). Thus, the small molecule inhibitor mirrored the effect observed when ERK dimerization was impeded by genetic means, behaving as a specific blocker of ERK extranuclear signals. As such, we used RSK1 phosphorylation as a readout to compare the inhibitory potentials of DEL-22379 and of SCH-772984, a recently described inhibitor of ERK kinase activity (Morris et al., 2013). Pretreating cells with either compound inhibited EGF-stimulated RSK1 phosphorylation, with estimated IC_{50} values of ~500 nM for both (Figure 2F). These data demonstrated that impeding ERK dimerization is as effective as inhibiting ERK catalytic activity to block biochemical effects elicited by ERK at the cytoplasm.

DEL-22379 Inhibits Growth of Tumor Cells Harboring RAS-ERK Pathway Oncogenes

Next, we investigated the biological effects of DEL-22379 on tumor cells in culture. We monitored a panel of human cell

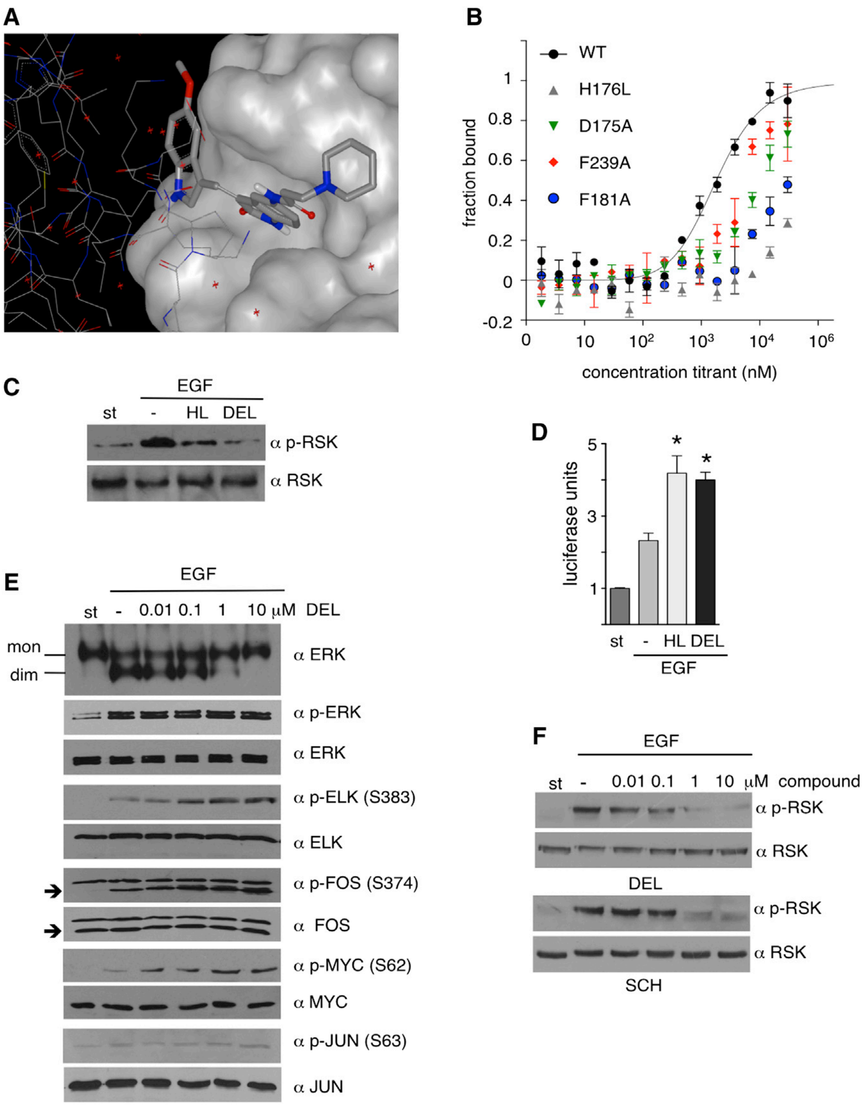

Figure 2. Effects of DEL-22379 on ERK Activity

(A) Docking of DEL-22379 on ERK dimerization interface as determined in silico.

(B) Binding of DEL-22379 to purified ERK2, WT or mutants within the binding groove, was evaluated by microscale thermophoresis. Data show the average ± SEM from two independent experiments.

(C and D) DEL-22379 effects on ERK cytoplasmic and nuclear functions. RSK1 phosphorylation (C) and ELK transactivation (D) in HEK293 cells treated with DEL-22379 (10 μM for 30 min) or transfected with ERK2 HL and then stimulated with EGF (100 ng/ml for 10 min). Data show the average ± SEM from three independent experiments. *p < 0.05 by unpaired t test.

(E) Effects of DEL-22379 on the phosphorylation of transcription factor substrates of ERK.

(F) Comparison of RSK1 phosphorylation inhibition by DEL-22379 and SCH-772984. (E) and (F) were assayed in EGF-stimulated HEK293 cells previously treated for 30 min with increasing concentrations of the indicated compounds.

See also Figure S2.

lines harboring mutant BRAF (V600E) or RAS (Q61L or G12V) from tumor types in which these mutations are prominent: melanoma (*BRAF* ~50%; *NRAS* ~15%) and colorectal cancer (*BRAF* ~20%; *KRAS* ~35%) (http://cancer.sanger.ac.uk/cancergenome/projects/cosmic/). In these, we compared the cytostatic effects of DEL-22379 to those of the MEK inhibitor PD-0325901 (Montagut and Settleman, 2009) and the ERK kinase inhibitor SCH-772984 (Morris et al., 2013), as reflected by their half-maximal growth inhibitory concentrations (GI_{50}). It was found that cell lines harboring mutant BRAF were the most sensitive to the three compounds. In comparison, wild-type (WT) cell lines for BRAF and RAS were the most resistant, and RAS mutant cells exhibited a range of sensitivities (Figure 3A). In cells showing different oncogenic genotypes, distinct sensitivity to DEL-22379 could not be attributed to variations on its effects on dimerization, because the molecule displayed similar dimerization- and cytoplasmic signaling-inhibitory dose responses (IC_{50} of 150–400 nM) regardless of the genotype (Figure 3B). To investigate the mechanism whereby the ERK dimerization inhibitor was affecting cellular viability, we analyzed its potential to induce apoptosis. It was significantly induced upon treatment with DEL-22379 in cell lines harboring oncogenic *RAS* or *BRAF*. However, in tumor cells devoid of such lesions, the apoptotic effects of the compound were attenuated (Figures 3C and 3D), similar to what has been shown for MEK and ERK kinase inhibitors (Morris et al., 2013; Solit et al., 2006), suggesting that the survival of these cells is largely independent of RAS-ERK signals. Thus, notwithstanding the differences in their dosages, quite expected when comparing pharmacologically optimized molecules like PD-0325901 and SCH-772984 to a "first hit" compound such as DEL-22379, these results demonstrate that MEK classical inhibitors, ERK kinase inhibitors, and ERK dimerization inhibitors exhibit similar therapeutic indexes with respect to the *RAS*/*BRAF* oncogenic signature, consistent with their mechanism of action affecting the same signaling pathway.

Noticeably, in cell lines with oncogenic *BRAF* or *RAS*, DEL-22379 GI_{50} values largely correlated with the ERK monomer-to-dimer ratio (Figure 4A), indicating a greater sensitivity to DEL-22379 in those cell lines where ERK dimerization was more prominent. It was of interest to gain an insight into what governed the degree of ERK dimerization. Because dimerization occurs mainly in the cytoplasm (Casar et al., 2008), we reasoned that factors regulating ERK nucleo- and cytoplasmic distribution could affect its dimerization. Accordingly, we found that the expression of PEA15, a protein that retains ERK at the cytoplasm (Formstecher et al., 2001), correlated with the degree of ERK dimerization present in BRAF mutant cell lines exhibiting similar

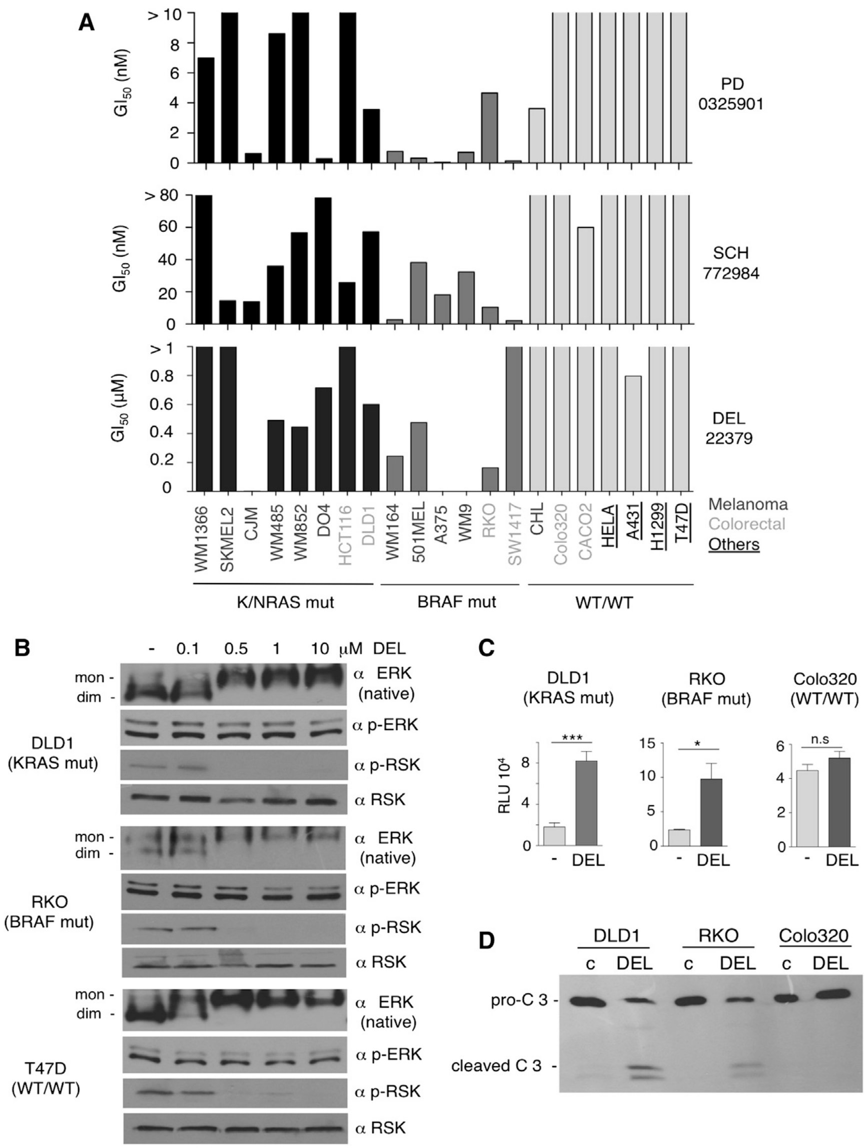

Figure 3. Evaluation of DEL-22379 Anti-proliferative Effects in Tumor Cells

(A) GI_{50} values for the indicated compounds in the indicated cell lines. Data show the average from five independent experiments.

(B) ERK dimerization inhibition by increasing concentrations (0.1–10 μM) of DEL-22379 in the indicated cell lines.

(C) Induction of apoptosis by DEL-22379 in the indicated tumor cell lines. Cells were treated with 10 μM DEL-22379 for 12 hr, and apoptosis was evaluated by Caspase-Glo 3/7 luminogenic assay. Data show the average ± SEM from five independent experiments. n.s, $p > 0.05$; *$p < 0.05$; ***$p < 0.005$ by unpaired t test; RLU, relative light units.

(D) As in (C), apoptosis was determined by western blotting for caspase 3 (C 3). The pro- and cleaved (active) forms of C 3 are indicated.

ERK phosphorylation levels (Figure 4B). Short hairpin RNA (shRNA)-mediated downregulation of PEA15 in A375 cells, harboring a high amount of dimeric ERK, markedly reduced the degree of ERK dimerization. Conversely, PEA15 overexpression in SW1417 cells, which displayed low ERK dimerization levels, resulted in a substantial increase on ERK dimers. In every case, enhanced dimerization resulted in potentiation of cytoplasmic signals, while attenuated dimerization favored nuclear signaling (Figure 4C).

In A375 cells, downregulation of PEA15 induced apoptosis (Figure 4D), probably as a consequence of the drop on the levels of ERK dimers, thereby resembling the effect of DEL-22379. Remarkably, the surviving cells, with reduced ERK dimerization levels, exhibited an augmented resistance to DEL-22379 compared to parental cells (Figure 4E). Conversely, in SW1417 cells, PEA15 overexpression, and the subsequent increment on ERK dimerization, increased their sensitivity to DEL-22379 (Figures 4E and 4F). These results show that PEA15 can markedly influence ERK dimerization and add further support to the notion that sensitivity to DEL-22379 in RAS-ERK pathway mutant cells is dictated by the levels of ERK dimerization.

DEL-22379 Prevents Tumor Growth and Metastasis in Animals

In light of the data obtained in cell cultures, we evaluated the anti-tumorigenic effects of the dimerization inhibitor in animal models. Prior to this, we analyzed DEL-22379 toxicity in mice. To determine a tolerable dose range for in vivo studies, mice were administered intra-peritoneal injections of DEL-22379 at doses ranging from 100 to 12 mg/kg (Figure S3A). The maximum tolerated dose was estimated at 15 mg/kg every 12 hr, defined as the highest dose that did not cause (1) alterations in motor coordination, activity, and muscular tone (Figure S3B) or (2) loss in body weight ≥ 20% (Figure S3C) or overt toxicity signs, such as abnormal behaviors, poor grooming, hunched posture, abnormal urine stains, or dehydration. In addition, the compound did not evoke toxicity in hematopoietic lineages (Table S2). In the liver, while some focal necrotic and steatotic lesions were evident (Figures S3D and S3E), the biochemical serum parameters for hepatocellular and cholestatic function were unaltered, indicating no gross liver malfunction (Table S3). In intestinal epithelia, shorter villi and mucus overproduction (Figures S3F and S3G) revealed some toxicity. Despite this, no diarrhea or overt signs of intestinal malfunction were evident. Thus, DEL-22379 toxicity should be considered mild. DEL-22379 pharmacokinetics was assessed in mice over a 72-hr time course following exposure to either single or multiple doses. Despite its short half-life in vivo, with a nearly complete clearance from plasma after the 12-hr dosing period, DEL-22379 plasma concentration reached a C_{max} of 1 μM at 1 hr, well over the IC_{50} values required to exert in vitro cytostatic effects in most

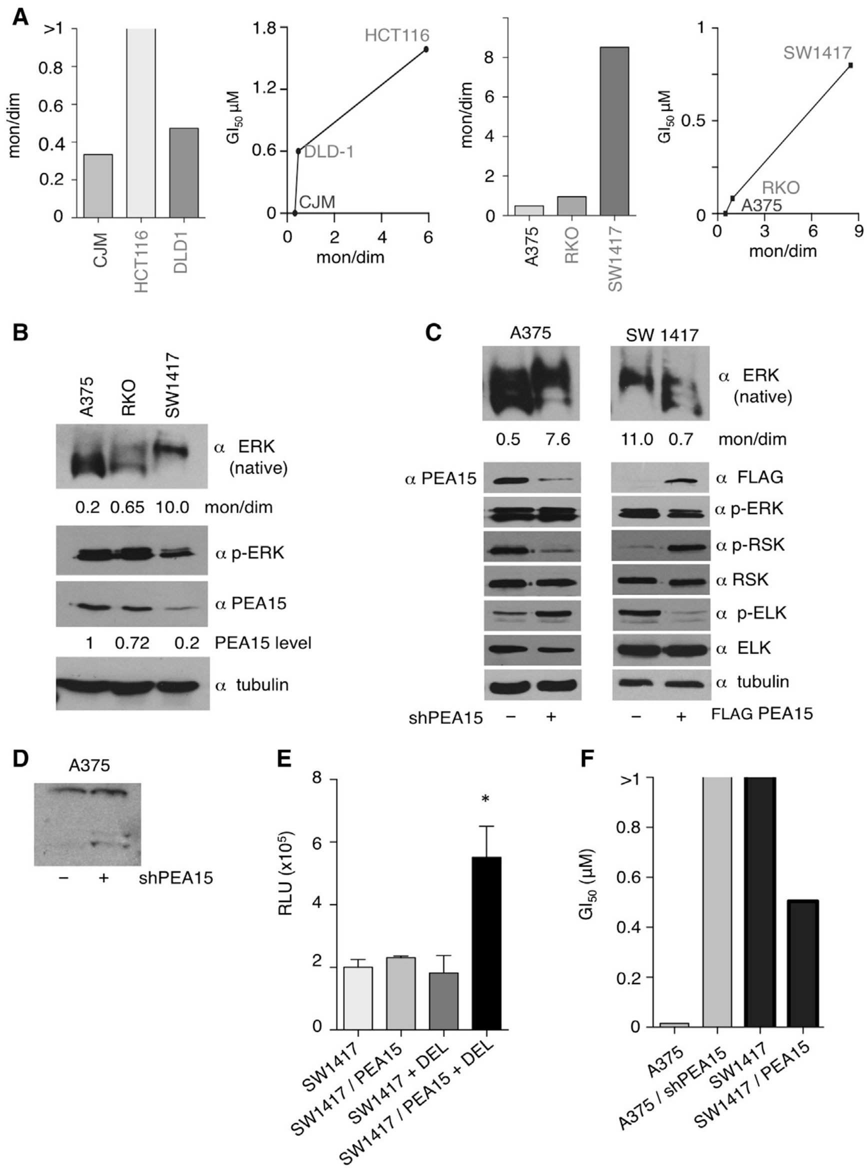

Figure 4. Regulation of ERK Dimerization by PEA15

(A) ERK monomers-to-dimers ratio in the indicated cell lines, and correlations with their respective GI_{50} values.

(B) Correlation of ERK dimerization levels with PEA15 levels in the indicated cell lines. Figures show the monomers-to-dimers ratio and PEA15 levels relative to the cell line with the highest levels.

(C) Impact of PEA15 on ERK dimerization and ERK nucleo- and cytoplasmic signals. The indicated cell lines were transfected with anti-PEA15 shRNA or FLAG-PEA15 (1 μg). Figures show the monomers-to-dimers ratio.

(D) Caspase 3 activation in A375 cells in response to downregulation of PEA15.

(E) Correlation between PEA15 levels and DEL-22379-induced caspase 3 activation in SW1417 cells. Where indicated, cells were treated with 1 μM of the compound for 12 hr, and caspase 3 activity was evaluated by Caspase-Glo 3/7 assay. Data show the average ± SEM from five independent experiments. *$p < 0.05$ by unpaired t test; RLU, relative light units.

(F) Correlation between PEA15 levels and DEL-22379 GI_{50} values. Data show the average from five independent experiments.

RAS-ERK pathway mutant cell lines, a concentration sustained over 8 hr (Figure S3H).

To test DEL-22379 antitumor effects, some of the aforementioned cell lines were xenografted into nude mice, and tumor growth was monitored after intra-peritoneal administration of the compound at 15 mg/kg. At such a dose, inhibition of ERK dimerization was evident in liver extracts (Figure 5A) and in xenografted tumors (Figure 5B). It was found that DEL-22379 markedly inhibited tumor progression for A375 cells (BRAF mutant) (Figures 5C and 5D). However, it failed to prevent the formation of tumors in mice injected with CHL cells (WT/WT) (Figures 5D and S3I). Surprisingly, tumors driven by activated RAS, like those derived from HCT116 cells, exhibited a pronounced sensitivity to DEL-22379 (Figures 5D and S3I), in contrast to their behavior in culture (Figure 3A). Interestingly, the proportion of ERK in dimeric form was much greater in HCT116-derived tumors than in HCT116 cells growing on plastic. Accordingly, cells from HCT116-derived tumors exhibited upregulated PEA15 levels and enhanced cytoplasmic signaling (Figure 5E). This suggested that for HCT116 cells, tumorigenesis in vivo had greater requirements for ERK dimerization than for proliferation in culture.

In light of DEL-22379 antitumor properties, we tested DEL-22379 efficiency in vivo using a colorectal cancer patient-derived xenograft (PDX) model that faithfully recapitulates the human disease (Puig et al., 2013). To this end, cells from *BRAF* mutant human colorectal adenocarcinoma were isolated as described (Puig et al., 2013) and injected subcutaneously into non-obese diabetic, severe combined immunodeficiency (NOD-SCID) mice. After detectable engraftment of the tumors (diameter > 0.4 mm), the evolution of the tumor mass was monitored in control and DEL-22379-treated mice. Remarkably, treatment with the dimerization inhibitor significantly mitigated growth of the *BRAF* mutant PDX (Figure 5F). Histopathological analyses of the tumors at the endpoint revealed that DEL-22379 caused extensive mucinous differentiation and cell death (Figure S3J), in agreement with those effects observed in both in vitro cultures and mouse xenografts using cancer cell lines.

To further substantiate our findings, we assayed whether DEL-22379 could revert the oncogenic process after the primary tumor had developed and metastatic dissemination was apparent. We utilized chicken embryo chorioallantoic membrane (CAM) xenografts (Deryugina and Quigley, 2008), which are especially suited for such analyses. Tumor cells were grafted on the

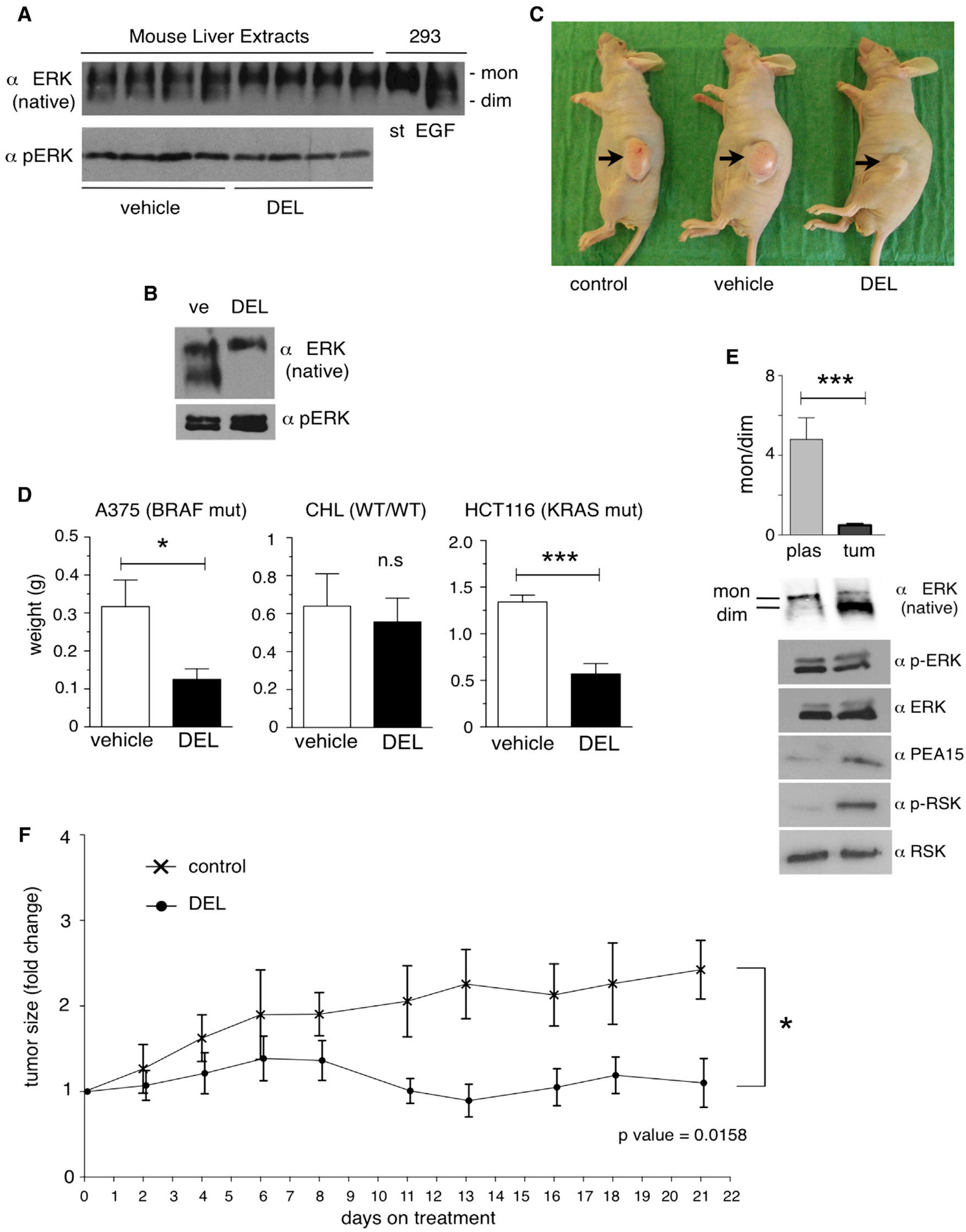

Figure 5. Antitumor Effects of DEL-22379 in Mice

(A and B) ERK dimerization in liver extracts (A) or tumors derived from HCT116 cells (B) from mice treated with DEL-22379 (15 mg/kg every 12 hr for 15 days). EGF-stimulated HEK293 cells are shown as the positive control. Ve, vehicle.

(C) Effects of DEL-22379 on the growth of tumors derived from A375 cells after 15 days.

(D) Correlation between DEL-22379 effects on tumor growth in mice and the mutational status of the xenografted cell lines.

(E) Comparison of the ERK monomers-to-dimers ratio in HCT116 cells derived from tumors in mice (tum) or growing on plastic plates (plas). In (D) and (E), data show the average ± SEM from five independent experiments. n.s., $p > 0.05$; *$p < 0.05$; ***$p < 0.005$ by unpaired t test.

(F) Effects of DEL-22379 on tumor growth in a *BRAF* mutant colorectal cancer PDX model. The longitudinal tumor growth plot shows the fold change of tumor volume for tumors formed in mice (n = 5–8) after 21 days of treatment. Error bars represent the ±SD p value by unpaired t test.

See also Figure S3 and Tables S2 and S3.

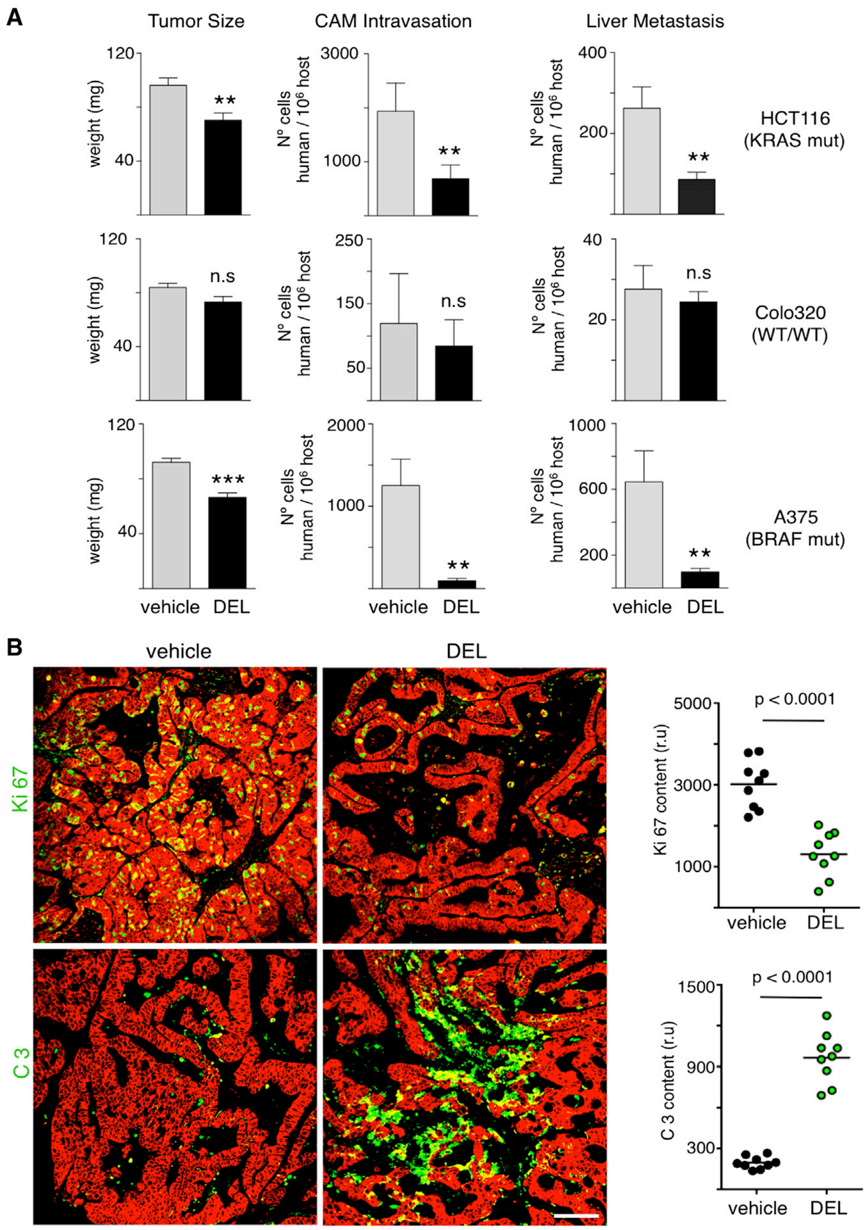

Figure 6. Effects of DEL-22379 on Tumor Progression and Metastasis

(A) DEL-22379 effects on tumor size, tumor cell intravasation, and liver metastases of the cell lines with indicated mutational status xenografted on chicken CAMs and allowed to progress for 5 days before drug administration (10 μM every 12 hr). Data show the average ± SEM from three independent experiments. n.s., $p > 0.05$; *$p < 0.05$; **$p < 0.01$; ***$p < 0.005$ by unpaired t test.

(B) Tumor regression in a *BRAF* mutant colorectal cancer orthotopic PDX model, treated with DEL-22379 (15 mg/kg every 12 hr) for 30 days. Representative pictures of immunofluorescence and confocal microscopy of sections from xenografts growing in the cecum wall. Ki67 and caspase 3 (C 3) expressions are shown in green; β-catenin (red) identifies cancer cells. Scale bar, 200 μm. Right panels: Column scatter plot for Ki67 or C 3 levels measured by immunofluorescence and confocal microscopy. Lines show mean values, p values by unpaired t test.

See also Figure S4.

embryo's CAM and allowed to expand for 5 days, resulting in extensive intravasation and liver metastasis. At this point, DEL-22379 was added and the evolution of the primary tumor and metastases was monitored. After 5 days, for A375 (BRAF mutant) and HCT116 (KRAS mutant) cells, a marked regression of the primary tumor size, as a consequence of prominent apoptosis (Figure S4), and drops in the number of cells in circulation and invading the liver were evident. Conversely, tumor progression in Colo320 (WT/WT) cells was undeterred (Figure 6A).

Further evidence on the ability of DEL-22379 to induce regression of advanced tumors was obtained from an orthotopic PDX model, in which *BRAF* mutant human colorectal adenocarcinoma cells were grafted in the cecum of NOD-SCID mice and allowed to progress for a month. At that point, DEL-22379 was administered for 30 days, and its antineoplastic effects were assessed. Treatment with DEL-22379 resulted in a pronounced arrest of tumor cell proliferation, as demonstrated by reduced Ki67 staining, accompanied by massive apoptosis, which was revealed by augmented cleaved caspase 3 levels (Figure 6B). Overall, these results demonstrated that targeting ERK dimerization pharmacologically prevented tumorigenesis at various stages when driven by RAS-ERK pathway oncogenes.

DEL-22379 Antineoplastic Effects Are Specifically Based upon Its Ability to Impede ERK Dimerization

In the course of the experiments dealing with CAM xenografts, we found that PD-0325901 and DEL-22379 reduced the growth of A375-derived tumors to similar extents (Figure 7A). However, PD-0325901 treatment resulted in a chicken embryo death rate of ~50%, probably as a consequence of potent ERK signaling suppression (Figure S5A). Conversely, embryos were unaffected by DEL-22379 (Figure 7A). Surprisingly, while ERK dimerization was apparent in the xenografted tumors, and markedly inhibited in those treated with DEL-22379, we could not detect ERK dimerization in the corresponding chick embryo livers in any case (Figure 7B), unlike what was previously observed in mouse livers (Figure 5A). Expectedly, phospho-ERK analyses in livers showed that *Gallus gallus* does not express ERK1 (Lefloch et al., 2008). The absence of ERK dimerization in chickens prompted us to interrogate other species across the evolutionary scale. We found that dimerization was also absent in ERK orthologs from amphibians (*Xenopus laevis*) and fish (*Danio rerio*) and only appeared in mammals: mice, dogs, and humans (Figure 7C). To further substantiate this point, we

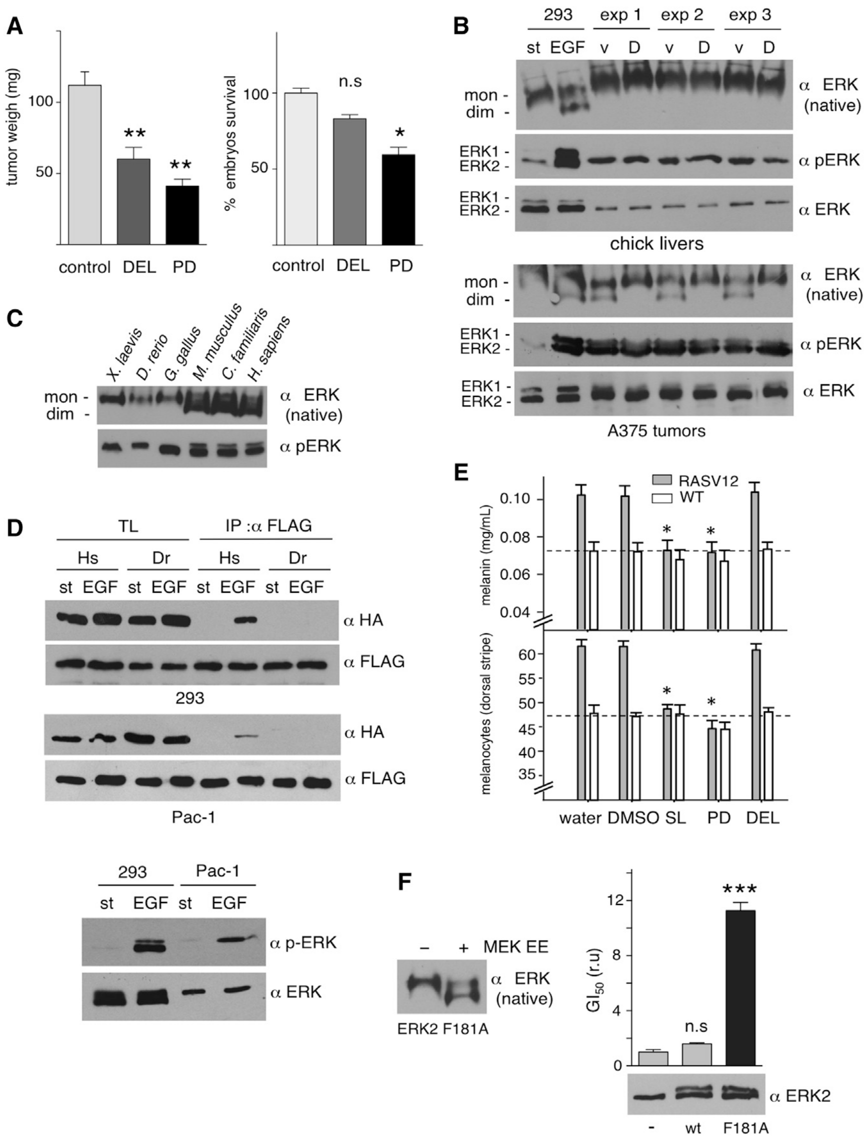

Figure 7. Association of DEL-22379 Antitumor Effects with Inhibition of ERK Dimerization

(A) Effects of DEL-22379 and PD-0325901 (10 μM each) on tumor growth (left) and chick embryo viability (right). Data show the average ± SEM from three independent experiments. n.s., $p > 0.05$; *$p < 0.05$; **$p < 0.01$ by unpaired t test.

(B) Effects of DEL-22379 on ERK dimerization in embryo livers (upper panels) and in A375-deriver tumors (lower panels). EGF-stimulated HEK293 cells are show as the control.

(C) Analysis of ERK dimerization in different animal species.

(D) HA- or FLAG-tagged versions of human (Hs) and zebrafish (Dr) ERK were co-transfected (1 μg each) in the indicated cell lines and tested for co-immunoprecipitation upon EGF stimulation (100 ng/ml for 10 min). St, starved cells. Lower panel: ERK phosphorylation in the indicated cell lines without or with EGF treatment.

(E) Melanin content and melanocytes numbers in WT and HRASV12 zebrafish larvae, treated with DEL-22379 or MEK inhibitors. Data show the average ± SEM. *$p < 0.05$ by ANOVA.

(F) DEL-22379 effects in the presence of ERK2 F181A. Left: dimerization of ERK2 F181A in vitro when incubated with activated GST-MEK ΔN EE. Right: DEL-22379 GI_{50} values in A375 cells overexpressing ERK2 F181A relative to those of parental cells. Data show the average ± SEM from three independent experiments. n.s., $p > 0.05$; ***$p < 0.001$.

See also Figure S5 and Tables S4 and S5.

co-expressed FLAG- and HA-tagged versions of zebrafish and human ERK2 in human (HEK293) and zebrafish (Pac-1) cells. Upon EGF stimulation, human HA-ERK2 co-immunoprecipitated with its FLAG-tagged version in both cell types, but we could not detect zebrafish HA-ERK2 associated with its FLAG-tagged form in any case (Figure 7D). These results ascertained that zebrafish ERK2 does not dimerize and suggest that such inability resides on the zebrafish protein, not on the cellular context.

This unexpected finding offered an invaluable scenario for testing DEL-22379 antitumor effects in the absence of ERK dimerization. Transgenic zebrafish expressing oncogenic *RASV12* in melanocytes develop melanomas, with neoplasia detectable at early larval stages (Michailidou et al., 2009) (Figures S5B–S5D). In these animals, melanoma development was characterized by increments in both melanocyte numbers and melanin contents with respect to WT fish. It was found that melanoma progression was prevented by treatment with the MEK inhibitors PD-184352 and SL-327. In contrast, DEL-22379 was ineffective in stopping neoplasia expansion up to the point when animals had to be sacrificed in accordance with humane endpoint guidelines (Figures 7E, S5E, and S5F).

These data suggested that DEL-22379 antineoplastic properties could be specifically attributed to the ability of DEL-22379 to prevent ERK dimerization. To further substantiate this point, we used ERK2 mutant F181A. This form is defective for binding to DEL-22379 (Figure 2B) yet retains its ability to dimerize (Figure 7F). When overexpressed in A375 cells, F181A conferred these cells with remarkable resistance to the inhibitor cytostatic effects, as shown by a dramatically increase in GI_{50} (Figure 7F). These results added support to the notion that in the absence of an ERK amenable to being targeted, DEL-22379 antitumor capacity is lost.

We investigated further DEL-22379 potential off-target effects by profiling the ability of DEL-22379 to inhibit a panel of 300 kinases. Noticeably, DEL-22379 displayed remarkable inactivity: the vast majority of the kinases tested were minimally affected, and only six kinases showed inhibition over 50% (Table S4). Of these, TRK-A, TRK-B, SNF1LK2, and MYLK2 are specifically expressed in tissues irrelevant to this study. STK17A is a pro-apoptotic kinase whose inhibition would potentiate, not inhibit, tumorigenesis (Mao et al., 2011). PDGFr-β expression was found

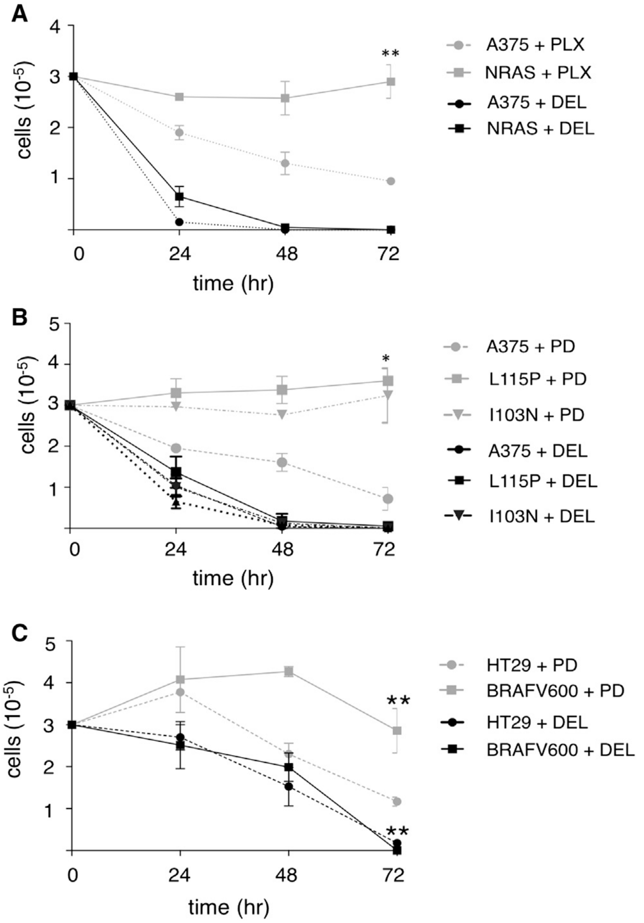

Figure 8. DEL-22379 Response to Classical Drug-Resistance Mechanisms

(A) Cell numbers of A375 cells, parental or overexpressing NRASV12, treated with DEL-22379 (1 μM) or the BRAF inhibitor PLX-4032 (10 μM) for the indicated periods of time.

(B) As in (A), cell numbers for A375 cells overexpressing the indicated MEK1 allosteric site mutants, treated with DEL-22379 or the MEK inhibitor PD-0325901 (10 μM).

(C) HT-29 cells (BRAF mutant), parental or overexpressing BRAFV600E, were treated with DEL-22379 or with PD-184352. In all cases, data show the average ± SEM from three independent experiments. *$p < 0.05$; **$p < 0.01$ by unpaired t test.

to be low or undetectable in the colorectal and melanoma cell lines used in our experiments (Nazarian et al., 2010) (Figure S5G). In these, we could not detect any relationship between PDGFr-β expression and sensitivity to DEL-22379.

Finally, we compared the transcriptional alterations evoked by an inhibitor that totally suppresses ERK signaling, such as the ERK kinase inhibitor SCH-772984, to those elicited by DEL-22379 that only suppress the ERK cytoplasmic signal. For this, A375 cells were treated with the aforementioned inhibitors for 2 hr, and the resulting transcriptomes were analyzed by RNA sequencing. SCH-772984 altered the expression of 505 genes. DEL-22379 altered the expression of only 34 genes, all of which were downregulated, including genes encoding proteins involved mainly in regulation of ERK activity (*DUSP6*, *SPRTY2*, and *SPRTY4*), survival (*IER3*, *MYC*, *DAPK3*, *SHARPIN*, and *PARP10*), vesicle trafficking (*RAB3*, *DPP7*, and *FASN*), blood vessel morphogenesis (*APOE*, *IL8*, and *ZFP36L1*), and nucleotide phosphate binding (*CHTF18*, *GALK1*, *NUBP2*, and *NAPRT1*). Noticeably, all but one (*ZFP36L1*) were in common with SCH-772984 (Table S5, the entire dataset is available at GSE68230). Altogether, the data mentioned earlier strongly indicate that the DEL-22379 therapeutic effect relies on its inhibition of ERK dimerization, not on unidentified off-target effects.

Inhibition of Tumor Cell Growth by DEL-22379 Is Unaffected by Drug-Resistance Mechanisms

Tumor cells acquire resistance to MEK and BRAF inhibitors by diverse molecular mechanisms (Little et al., 2013), so it was important to determine whether such processes affected DEL-22379. In melanomas, upregulation of NRAS confers resistance to BRAF inhibitors in BRAF mutant tumors (Nazarian et al., 2010). In agreement, we found that overexpression of NRASV12 in A375 cells made them refractory to the anti-proliferative effect of the BRAF inhibitor PLX-4032. Contrarily, these cells remained sensitive to inhibition by DEL-22379 (Figure 8A). Similarly, point mutations that arise in the allosteric drug-binding pocket of MEK1 (Emery et al., 2009) result in melanoma resistance to MEK inhibitors. We simulated such situation by the overexpression of MEK1 mutants L115P and I103N (Emery et al., 2009) in A375 cells, which made them insensitive to proliferative arrest by the MEK inhibitor PD-0325901, but this could not revert growth inhibition in response to DEL-22379 treatment (Figure 8B). Likewise, amplification of oncogenic *BRAF* in colorectal cancer underpins acquired resistance to drugs inhibiting MEK (Little et al., 2011). As such, we found that overexpression of BRAFV600E in HT29 cells made them unresponsive to treatment with PD-0325901, while they remained entirely sensitive to DEL-22379 (Figure 8C). Overall, these results demonstrated that the cytostatic effect of the ERK dimerization inhibitor DEL-22379 is indifferent to drug-resistance mechanisms that hamper classical RAS-ERK pathway inhibitors.

DISCUSSION

In our search for mechanisms of action to be exploited as therapeutic targets for neoplasia driven by RAS-ERK pathway oncogenes, we have focused on ERK dimerization. We had previously unveiled this regulatory protein-protein interaction as a potential target for antitumor drugs, since preventing it by molecular biology approaches forestalled tumor progression for cell lines harboring oncogenic *KRAS* (Casar et al., 2008). In order to screen for compounds capable of inhibiting ERK dimerization, we set up a method for detecting ERK dimerization using native gel electrophoresis and extensively validated it. Using this technique, we identified the 3-arylidene-2-oxindole derivative DEL-22379. By diverse methodologies we verified that DEL-22379 is a bona fide ERK dimerization inhibitor that does not alter ERK phosphorylation levels and, consequently, does not affect ERK kinase activity. We identified the DEL-22379 binding site on ERK2 in a cleft within its dimerization interface, close to residues that play essential roles in the dimerization process, such as His 176 (Wilsbacher et al., 2006). Thus, we can speculate that DEL-22379 inhibits ERK dimerization by docking onto the dimerization interface, thereby preventing the interaction between two ERK molecules taking place.

We have extensively studied DEL-22379 antitumor properties in diverse models for RAS-ERK pathway-driven neoplasia. It has been demonstrated that the anti-proliferative efficacy of kinase inhibitors aimed at RAS-ERK signals depends on the driver oncogenic lesion (Bollag et al., 2010; Morris et al., 2013; Poulikakos et al., 2010; Solit et al., 2006). We found that DEL-22379 exhibits the greatest cytostatic potential against tumor cell lines harboring mutant BRAF; it has variable effects in those with oncogenic RAS, and it is largely ineffective in cell lines devoid of RAS-ERK pathway mutations. Thus, DEL-22379 fully resembles MEK (Solit et al., 2006) and ERK (Morris et al., 2013) kinase inhibitors. These data suggest that the DEL-22379 therapeutic effect is specifically mediated through its impact on ERK by preventing its dimerization. This notion is further endorsed by our results showing that the sensitivity of tumor cell lines to DEL-22379 correlates greatly with their content of ERK in dimeric form. ERK dimerization is known to depend on ERK phosphorylation (Khokhlatchev et al., 1998) and to occur in the presence of cytoplasmic components (Philipova and Whitaker, 2005), such as scaffold proteins (Casar et al., 2008). Adding to this, we now unveil PEA15 as a promoter of ERK dimerization, probably by retaining ERK at the cytoplasm, thereby facilitating its interaction with scaffolds therein. Therefore, differences in these factors, and probably in others hitherto unidentified, among distinct cellular contexts may alter the degree of ERK dimerization and, consequently, the response to DEL-22379.

More importantly, our data demonstrate that DEL-22379 is effective as an antitumor agent in animal xenograft models for RAS-ERK pathway-driven carcinogenesis, with low toxicity at therapeutic doses. In this realm, DEL-22379 elicits a pronounced apoptotic response in xenografts from melanoma and colorectal cancer cell lines harboring mutant *BRAF* and *RAS*. By this mechanism, DEL-22379 is capable not only of preventing the growth of nascent tumors but also of reducing primary tumor mass and metastatic dissemination in tumors allowed to progress prior to treatment with the compound. Of particular importance are our data showing that DEL-22379 is effective in preclinical orthotopic PDX models for *BRAF* mutant colorectal cancer. At present, there is no effective, lasting drug therapy against this type of colorectal cancer (Holderfield et al., 2014). Therefore, ERK dimerization inhibitors may open a promising venue for future therapies.

By different means, we have demonstrated that DEL-22379 antineoplastic effects are specifically based on its properties as an inhibitor of ERK dimerization and are not a consequence of unidentified off-target effects. Importantly, we show that DEL-22379 is ineffective for preventing melanoma progression in zebrafish, a species in which ERK does not form dimers. The absence of ERK dimerization in some species is a surprising finding currently under study in our laboratory. Interestingly, we show that DEL-22379 alters the expression of only 34 genes, all but one in common with those affected by the ERK kinase inhibitor SCH-772984, further indicating its specificity for ERK signaling. This result also illustrates that DEL-22379 has few transcriptional effects, in full agreement with its function as an inhibitor of ERK extranuclear signals, and that its pro-apoptotic functions must be exerted largely by transcription-independent mechanisms. Moreover, this result implies that the increased phosphorylation of exchange factors such as ELK, FOS, and MYC observed upon DEL-22379 treatment would have few transcriptional consequences, probably because ERK-mediated transcriptional output would be at its maximum prior to the nuclear influx of additional ERK monomers resulting from the disruption of ERK dimers by DEL-22379. Interestingly, it has been reported that knockdown of the scaffold protein KSR1, thereby downregulating ERK cytoplasmic signaling, impairs MYC oncogenic function and enhances MYC-mediated apoptosis (Gramling and Eischen, 2012). It is tempting to speculate that enhanced MYC phosphorylation following DEL-22379 treatment serves this purpose.

We had previously described ERK dimers as playing an essential role in the regulation of ERK cytoplasmic signals but not nuclear ones and how inhibiting the ERK extranuclear component was sufficient to forestall cellular transformation and tumor progression (Casar et al., 2008). DEL-22379 fully recapitulates these notions. In agreement, melanoma response to the BRAF inhibitor PLX-4032 correlates with a drop in cytoplasmic, but not nuclear, ERK activity (Bollag et al., 2010). Thus, our findings show that a small molecule inhibitor specifically targeting the extranuclear component of RAS-ERK signals can be as effective as inhibitors aimed at completely blocking RAS-ERK flux for antineoplastic intervention.

Overall, we demonstrate that a molecule inhibiting a regulatory protein-protein interaction such as ERK dimerization can be effectively used in cancer therapy. Our results endorse recent findings pointing to other regulatory protein-protein interactions, such as dimerization of BRAF (Freeman et al., 2013) and ARAF (Mooz et al., 2014), as potential therapeutic targets in the RAS-ERK pathway. Indeed, this strategy has proven successful in other realms. For example, drug-mediated inhibition of the interaction between the leukemia-associated proteins RUNX1 and CBF-β prevented proliferation of leukemic cells (Gorczynski et al., 2007).

Our findings demonstrate that inhibiting ERK dimerization by a compound that prevents ERK-ERK interaction evokes antineoplastic effects. This opens the possibility that interfering with other mechanisms involved in ERK dimerization would yield similar therapeutic outcomes. ERK scaffold proteins serve as dimerization platforms (Casar et al., 2008). Thus, small molecules aimed at preventing ERK-scaffold interactions could pose an alternative strategy for inhibiting ERK dimerization and thereby a therapeutic venue for antineoplastic intervention. This notion is supported by findings demonstrating that disruption of scaffold proteins like KSR1 (Lozano et al., 2003) and IQGAP1 (Jameson et al., 2013) can effectively prevent RAS-driven tumorigenesis.

In addition, our ERK dimerization inhibitor can overcome molecular processes that confer resistance to BRAF and MEK inhibitors. Furthermore, ERK dimerization inhibitors could prove useful in situations in which ATP-competitive ERK inhibitors fail, like those cases in which the appearance of ERK mutations confer resistance to such inhibitors by interfering with drug binding (Goetz et al., 2014). Thus, these kinds of compounds could provide efficacious second-wave therapeutics for those cases in which resistance to classical RAS-ERK pathway inhibitors emerges. Moreover, it is conceivable that ERK kinase inhibitors would be insensitive to hypothetical ERK dimerization interface mutations that would inactivate dimerization inhibitors. Therefore, both types of ERK inhibitors could replace each other, depending on the resistance mechanism to be treated. In addition, though purely speculative at this stage, since ERK dimerization inhibitors can produce a therapeutic effect by just partial suppression of ERK functions, it is possible that they cause fewer undesired side effects than those therapeutics that suppress ERK activation completely. Indeed, the mild toxicity exhibited in mice suggests this effect. Thus, this may open the door for higher dosages and prolonged treatments, evoking improved cancer treatments.

EXPERIMENTAL PROCEDURES

Native PAGE for detecting ERK dimers was performed exactly as previously described (Pinto and Crespo, 2010). Compound screening was performed in HEK293T cells treated with the potential inhibitors (10 μM) for 30 min before EGF stimulation. Cellular lysates were tested for ERK dimerization by native PAGE and p-ERK evaluation of the potential positives.

In vitro ERK dimerization was assayed using GST-MEK1 ΔN EE purified from bacteria, bound to glutathione sepharose beads, and incubated with 25 μg/ml of purified His-ERK2 plus increasing concentrations of DEL-22379. Western blotting, kinase assays, and luciferase assays also performed as described (Casar et al., 2008). In silico docking of the DEL-22379 compound was carried out with the modeling tools provided by the OpenEye package (v. 2.1) (http://www.eyesopen.com).

Tumor cell lines were from our collection or from ATCC. GI_{50} were analyzed as described (Solit et al., 2006): cells were plated at a density of 1,000–2,000 cells/well and treated with the drugs for 48 hr, Alamar Blue (Invitrogen) was added, and the colorimetric change was measured at 570 and 600 nm. GI_{50} was estimated by nonlinear regression using GraphPad5 Prism Software. Apoptosis was analyzed by evaluating caspase 3 activity, either by western blotting or using the Caspase-Glo 3/7 luminogenic assay (Promega).

Cancer cells were xenografted in female, athymic nu/nu mice of 8 weeks of age as described (Casar et al., 2008). 3×10^6 cells were injected subcutaneously in the lateral flank and allowed to develop for 10–15 days before treatment with DEL-22379 at 15 mg/kg every 12 hr for 2 weeks. PDXs were performed using patient-derived colorectal cancer cells harboring BRAFV600E from non-necrotic areas of primary adenocarcinomas from patients that underwent surgical resection, basically as described (Tenbaum et al., 2012). Written informed consent was signed by all patients. The project was approved by the Research Ethics Committee of the Vall d'Hebron University Hospital, Barcelona, Spain (Approval ID: PR(IR)79/2009). Cells were grafted in both flanks or in the cecum of NOD-SCID mice. DEL-22379 was administered by intra-peritoneal injection at a concentration of 15 mg/kg every 12 hr for 30 days.

Assays to evaluate metastasis in chick embryos were performed as described (Deryugina and Quigley, 2008). We inoculated 10^6 tumor cells on the chorioallantoic membrane and incubated the eggs for 5 days at 38°C. Then, the different drugs were added topically on the upper CAM; treatments were refreshed every 12 hr. Human cells within chick embryo tissues were detected by real-time Alu PCR.

Carcinogenesis experiments in zebrafish were carried out as described (Michailidou et al., 2009). Treatments with inhibitors were performed in larvae at 2 days post-fertilization (dpf). Larvae were incubated in the presence of inhibitors for 72 hr, refreshed every 24 hr. At 5 dpf, the experiments were terminated.

All experiments involving animals were approved by the Animal Care Committee of the University of Cantabria to verify compliance with the Spanish (Real Decreto 53/2013) and European Communities Council Directive 2010/63/UE, "Protection of Animals Used in Experimental and Other Scientific Purposes." Detailed descriptions of the different protocols are available as Supplemental Experimental Procedures.

ACCESSION NUMBERS

The National Center for Biotechnology Information's Gene Expression Omnibus database accession number for the microarray datasets is GSE68230.

SUPPLEMENTAL INFORMATION

Supplemental Information includes Supplemental Experimental Procedures, five figures, and five tables and can be found with this article online at http://dx.doi.org/10.1016/j.ccell.2015.07.001.

AUTHOR CONTRIBUTIONS

DEL-22379 was synthesized at Vichem by Z.H. under the supervision of L.O. and G.K. P.C.-B. and B.C. performed most of the experiments, with the exception of those performed by A.H. (Figures 1D, 3A, and 3C), and Figures 1A and 5A and the drug screening, which were performed by A.P. Experiments using mice and toxicology studies were performed by R.V. and E.M.V. and supervised by A.P. S.P.T. performed the PDX experiments, supervised by H.G.P. L.A.-I. performed the PLAs. R.G. performed the in silico docking. P.N. performed the histopathological analyses. E.B. performed the pharmacokinetic determinations. M.J. performed the zebrafish experiments, supervised by A.H. G.R. performed analytical centrifugations. A.P.-L. performed ERK docking experiments. A.H. and B.C. also prepared the figures and performed the statistical analyses. P.C. conceived the study, directed it, analyzed and interpreted data, and wrote the manuscript.

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